

Harnessing Big Data Tools in Financial Services

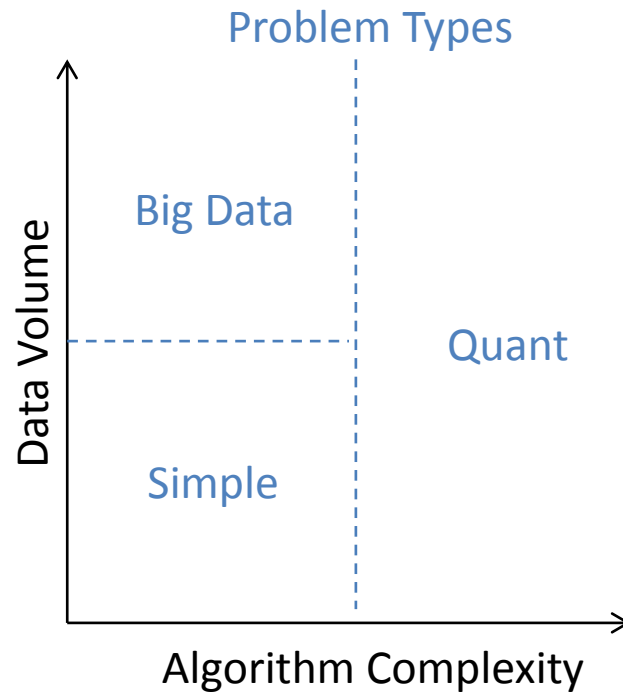
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Big Data – a little analysis

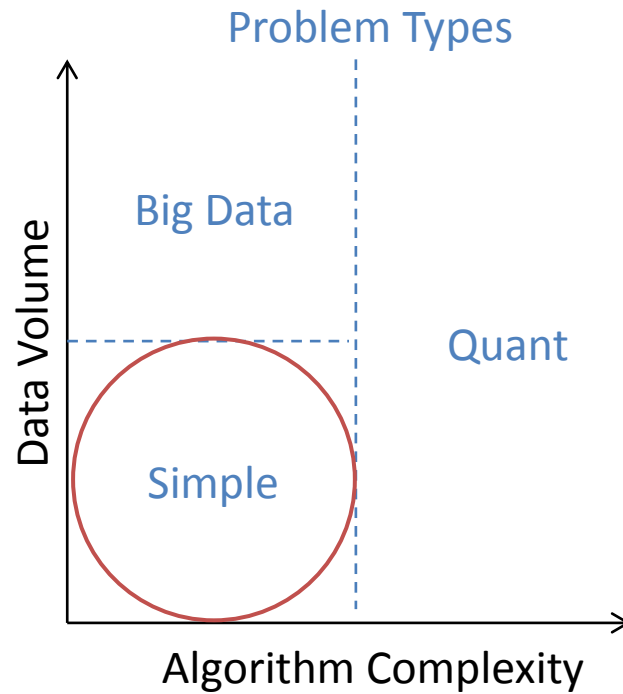
Overview

Based on a blog post from April 2012 – <http://is.gd/swbdla>



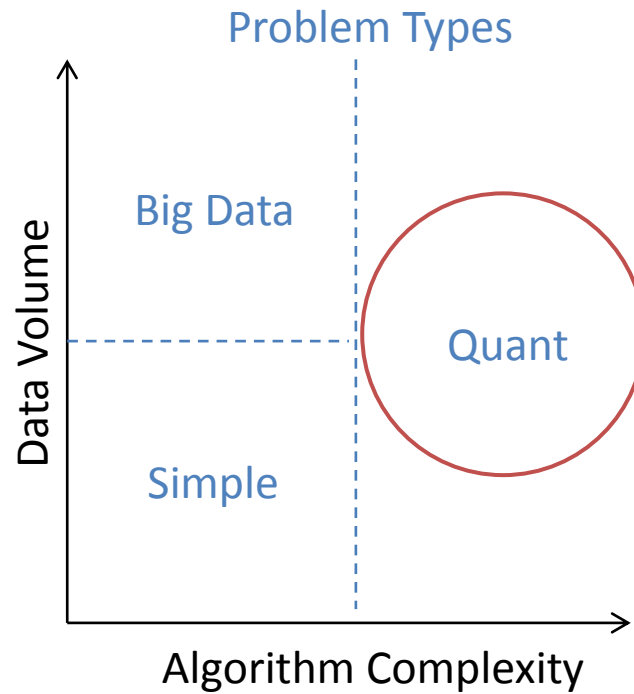
Simple problems

Low data volume, low algorithm complexity



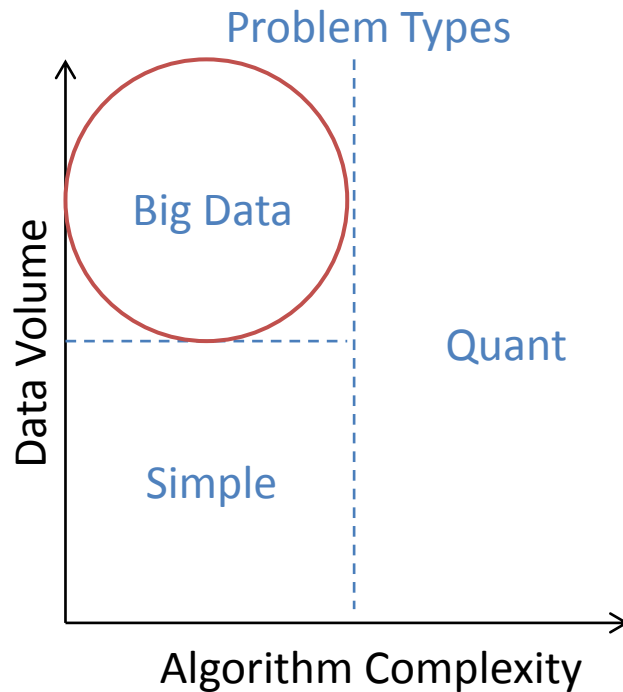
Quant Problems

Any data volume, high algorithm complexity



Big Data Problems

High data volume, low algorithm complexity



Types of Big Data Problem:

1. Inherent
2. More data gives better result than more complex algorithm

The good, the bad and the ugly of Big Data

Good

- Lots of new tools, mostly open source



Bad

- Term being abused by marketing departments

Ugly

- Can easily lead to over reliance on systems that lack transparency and ignore specific data points
 - 'Computer says no', but nobody can explain why

*Whoever thinks their analytics problem is
solved by big data,
doesn't understand their analytics
problem and doesn't understand
big data*



Security and Governance

The priesthood of storage and the cult of the DBA

- Enterprise storage systems have (mostly) their own interconnect and their own special people to look after that, any changes (weekends only) and backups
 - The priesthood of storage
- Relational Database Management Systems (RDBMS) are about more than just SQL
 - Backup and recovery
 - Access control
 - Identity management
 - Integration with enterprise directories
 - Data security
 - Encryption
 - Schema management
 - Glossaries and data dictionaries
- DataBase Administrators (DBAs) have become the guardians of all this
 - The cult of the DBA
- Anything not under the management of the cult doesn't count as being part of the official 'books and records of the firm'
 - Or at least that's what they'll tell you



NOSQL as a hack around corporate governance

Many 'Big Data' tools also fly under the banner of 'NOSQL'

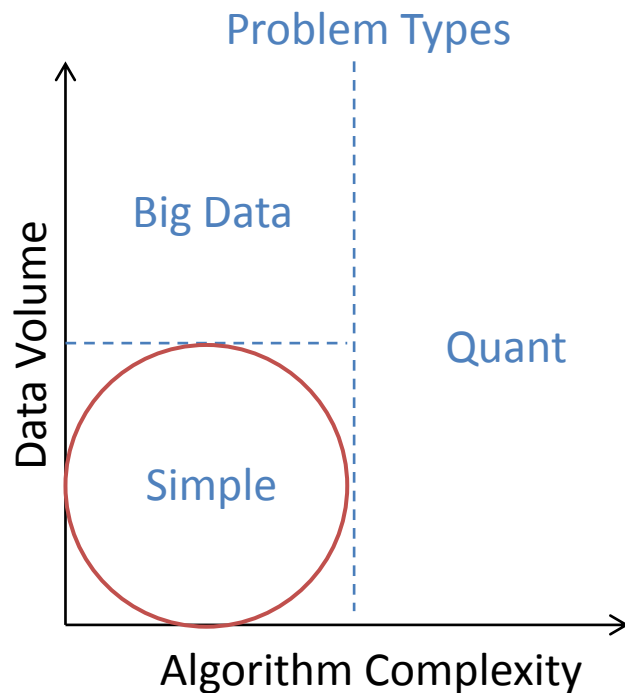
NOSQL allows for the escape from the clutches of the priesthood of storage and the cult of the DBA

- The reason for choosing Cassandra (or whatever) for a project might have nothing to do with 'Big Data'
- Security is often viewed as an optional non functional requirement
 - Big Data security controls may be less mature than traditional RDBMS
 - So compensating controls must be used for whatever is missing out of the box
 - 3rd party tools market still nascent
 - So less choice for bolt on security
- NOSQL hasn't yet become an integral part of organisation structure/culture

Data Centre implications

Simple problems

Low data volume, low algorithm complexity

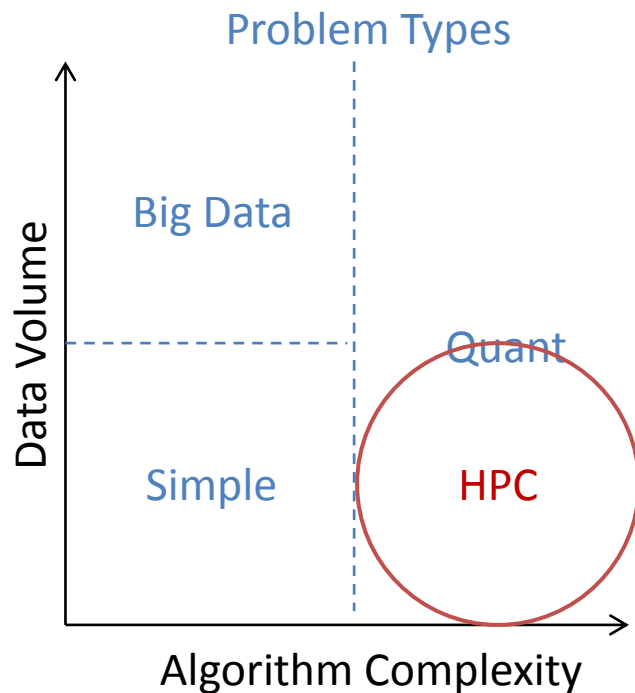


This is the type of problem that has traditionally worked a single machine (the database server) really hard.

- Reliability has always been a concern for single box designs (though this is a solved problem where synchronous replication is used).
 - This is what makes SAN attractive
- No special considerations for network and storage

Quant Problems – the easy part

Any data volume, high algorithm complexity

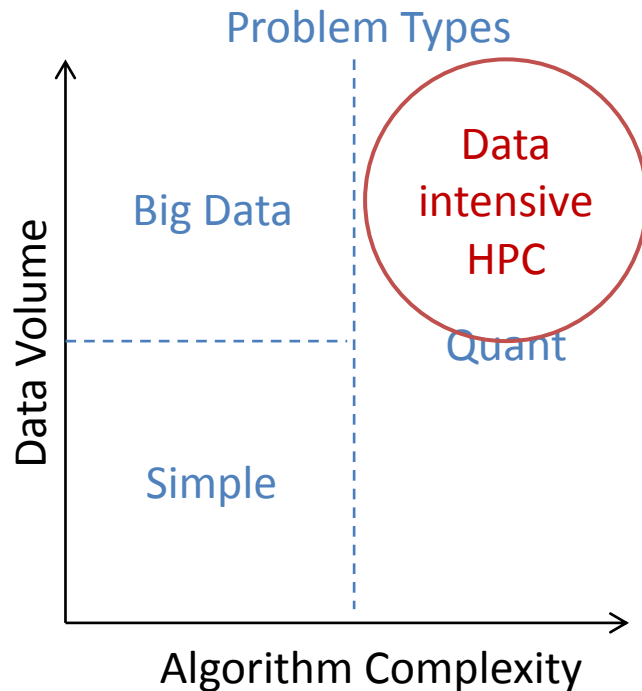


High Performance Compute (HPC) impact is well understood:

- Lots of machines at the optimum CPU/\$ price point
 - Previously optimised for CAPEX
 - Present trend is to optimise for TCO (especially energy)
- No real challenges around storage or interconnect
 - Though some local caching using a 'data grid' may improve duty cycle over a pure stateless design

Quant Problems – the hard part

Any data volume, high algorithm complexity

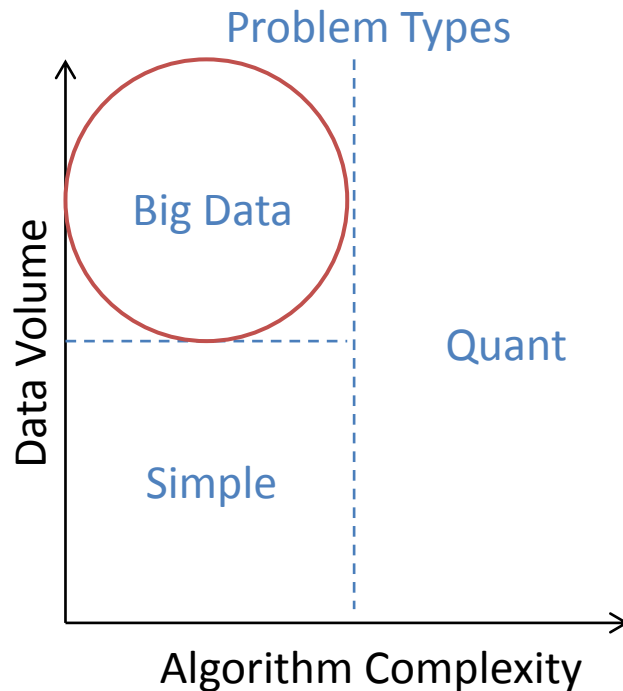


Data intensive HPC shifts the focus to interconnect and storage:

- Fast network (>1gB Ethernet) may be needed to get data where it's needed
 - 10gB Ethernet (or faster)
 - Infiniband if latency is an issue
- SANs don't work at this scale (and are too expensive anyway)
 - Data needs to be sharded across inexpensive local discs

Big Data Problems – look easy now

High data volume, low algorithm complexity



Typically less demanding on interconnect than data intensive HPC workloads:

- Ethernet likely to be sufficient
- Many things that wear the 'big data' label are in fact solutions for sharding large data sets across inexpensive local disc
- E.g. This is what the Hadoop Distributed File System (HDFS) does

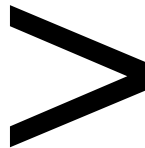
The role of SSD

- At least for the time being this is a delicate balance between capacity and speed
- Applications that become I/O bound with traditional disc need to make a value judgement on scaling the storage element (switch to SSD) versus scaling the entire solution (buy more servers and electricity).
 - Falling prices will tilt balance towards SSD
- Worth noting that many traditional databases will now fit into RAM (especially if spread across a number of machines), which leaves an emerging SSD sweet spot across the middle of the chart.
- Attention needs to be paid to the 'impedance mismatch' between contemporary workloads (like Cassandra) and contemporary storage (like SSD). This is not handled well by decades old file systems (and for a long time the RDBMS vendors have cheated by having their own file systems).
- SSD will hit the feature size scaling wall at the same time as CPU
 - Spinning disc (and other technologies will not)
 - Enjoy the ride whilst it lasts (perhaps not too much longer)
 - Interesting things will happen when things we've become accustomed to having exponential growth flatten out whilst other growth curves continue

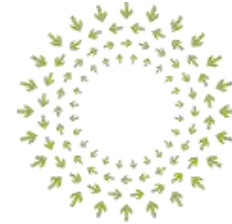
The future of block storage

- SAN/NAS stops being a category in its own right and becomes part of the software defined data centre
 - SAN (and especially dedicated fibre channel networks) goes away altogether
 - NAS folds into the commodity server space – looks like DAS at the hardware layer but behaves like NAS from a software perspective
 - Dedicated puddles of software defined storage will be aligned to 'big data', but the overall capacity management should ultimately be defined by the first exhausted commodity (CPU, RAM, I/O, disc)

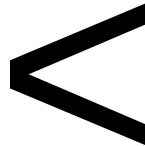
Data Centre impact - Summary



Simple energy efficient servers
With local disk



OPEN
Compute Project



Big boxes
Connected to SAN

Everything looks the same (less diversity in hardware)

Everything uses the minimum possible energy

'Big Data' is a part of the overall capacity management problem

Data centre automation will solve for optimal equipment/energy use

Wrapping up

Conclusions

- 'Big Data' is a label that used to describe an emerging category of tools that are useful for problems with large data volume and low algorithmic complexity
- The technical and organisational means to provide security and governance for these tools are less mature than for traditional databases
- Data centres will fill up with more low end servers using local storage (and these will likely be the designs emerging from hyperscale operators that are optimised for manufacturing and energy efficiency)

Questions?